

Recent Advances in Deep Learning - Spring 2026

Final Examination (Mock Test C) - Answer Key & Rubric

Total Marks: 100

Part I: True/False Questions (10 questions, 1 mark each, Total: 10 marks)

1. ✗ (ReLU does not "completely" solve the problem for all inputs. For inputs less than zero, the gradient is exactly zero, which can lead to the "Dying ReLU" phenomenon.)
2. ✓ (Max-pooling operations provide local translation invariance by selecting the maximum activation regardless of slight shifts in the input.)
3. ✓ (During inference, Dropout is turned off to use the full capacity of the network, and weights are scaled accordingly—or inverted dropout is used during training.)
4. ✓ (The self-attention mechanism computes the relationships between all tokens simultaneously, breaking the sequential computation bottleneck of RNNs.)
5. ✓ (L2 regularization penalizes large squared weights, encouraging the network to distribute weights more evenly rather than relying on a single feature.)
6. ✓ (This is the fundamental definition of Batch Normalization's standardization step.)
7. ✗ (A policy function $\pi(a|s)$ maps a state to a probability distribution over actions. It is the Value Function $V(s)$ or $Q(s, a)$ that estimates the expected return.)
8. ✓ (Adam combines the benefits of both Momentum (first moment/gradient EMA) and RMSProp (second moment/squared gradient EMA).)
9. ✗ (An underfitted model has *high* training error because it fails to capture the underlying patterns in the data.)
10. ✓ (In order to perform the dot product QK^T , the feature dimension d_k of the Query and Key matrices must be identical.)

Part II: Multiple Choice Questions (20 questions, 2 marks each, Total: 40 marks)

1. **B. To allow the network to learn and represent non-linear decision boundaries.**
(Without non-linear activations, a multi-layer network collapses mathematically into a single linear transformation.)
2. **B. Softmax and Cross-Entropy Loss**

3. **C. The loss may fluctuate drastically and even diverge.**
4. **C. They act as a channel-wise linear transformation and can be used for dimensionality reduction.** (Often used as a "bottleneck" layer to reduce computational cost by decreasing the number of channels.)
5. **C. LSTM introduces a cell state and gating mechanisms that mitigate the vanishing gradient problem for long sequences.**
6. **B. To inject information about the relative or absolute position of the tokens in the sequence.**
7. **B. Increase the size and diversity of the training set to prevent overfitting.**
8. **B. Value-based methods learn the optimal action-value function (Q -function), while Policy Gradients directly optimize the policy $\pi(a|s)$.**
9. **A. L1 Regularization (Lasso)**
10. **C. By dividing it into non-overlapping patches, and linearly projecting each flattened patch.**
11. **C. Masked Multi-Head Attention** (The mask prevents the model from "looking ahead" at future tokens, preserving the autoregressive property.)
12. **B. Underfitting the training data.** (High bias means the model is too simple; low variance means its predictions do not change much across different training sets.)
13. **B. To estimate the value function and evaluate the action chosen by the Actor.**
14. **C. $O(N^2)$** (Calculating the attention matrix requires computing the dot product between every pair of tokens in a sequence of length N .)
15. **B. The source task and the target task share low-level features (e.g., edges, shapes in images).**
16. **B. Clipping the probability ratio to prevent the new policy from deviating too far from the old policy.**
17. **B. Degradation problem (training error increasing as network gets deeper)** (It also effectively alleviates vanishing gradients.)
18. **B. The region in the input space that a particular CNN feature is looking at.**
19. **B. Feeding the ground truth token from the previous time step as input to the decoder, instead of the model's own prediction.**
20. **B. F1-Score (or Precision/Recall)** (Accuracy is highly misleading on imbalanced datasets; e.g., predicting all instances as the majority class yields 99% accuracy but is a useless model.)

Part III: Short Answer & Essay Questions (Open-end, 5 questions, 10 marks each, Total: 50 marks)

1. Vanishing Gradients and Architecture Solutions (10 marks)

- **(1) Mathematical Reason (5 marks):**

By the chain rule, gradients in deep networks are products of derivatives of the activation functions and weights. The maximum derivative of the Sigmoid function is 0.25. Multiplying many values less than 1 causes the gradient to decay exponentially as it propagates backward, resulting in near-zero gradients at shallow layers.

- **(2) Architecture Solutions (5 marks):**

ReLU: For positive inputs ($x > 0$), the derivative of ReLU is exactly 1. This prevents the gradient from decaying as it passes through the activation function.

ResNet (Skip Connections): By defining $H(x) = F(x) + x$, the derivative becomes $\frac{\partial H}{\partial x} = \frac{\partial F}{\partial x} + 1$. The "+1" term acts as a gradient highway, ensuring that even if $\frac{\partial F}{\partial x}$ vanishes, the gradient can still flow unattenuated to earlier layers via the identity mapping.

2. Generalization and Regularization (10 marks)

- **(1) Bias-Variance Tradeoff (4 marks):**

Overfitting: Characterized by **Low Bias** and **High Variance**. The model is overly complex and memorizes noise in the training data, leading to poor generalization on unseen data.

Underfitting: Characterized by **High Bias** and **Low Variance**. The model is too simple to capture the underlying patterns, performing poorly on both training and test sets.

- **(2) Regularization Techniques (Select two, 3 marks each, Total 6 marks):**

Dropout: Randomly zeroes out a fraction of neurons during training. This prevents complex co-adaptations between neurons, forcing the network to learn robust, independent features (acting like an implicit ensemble).

L2 Regularization (Weight Decay): Adds a penalty term proportional to the sum of squared weights to the loss function. It forces weights to be smaller and more evenly distributed, resulting in a smoother function less sensitive to individual input variations.

Early Stopping: Monitors validation loss during training. Training is halted when validation loss stops decreasing and begins to rise (the onset of overfitting), saving the weights that generalize best.

3. The Transformer Core Mechanisms (10 marks)

- **(1) Mathematical Formula (3 marks):**

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

- **(2) Multi-Head Advantage (4 marks):**

A single attention head might focus on a single aspect of the sequence. Multi-Head Attention linearly projects Q, K, V into multiple representation subspaces, allowing the model to jointly attend to information from different positions and linguistic/semantic aspects in parallel (e.g., one head for syntax, another for semantic entities).

- **(3) Scaling Factor $\frac{1}{\sqrt{d_k}}$ (3 marks):**

For large values of d_k , the dot product QK^T can grow extremely large in magnitude, pushing the Softmax function into regions where gradients are extremely small (saturation). Scaling by $\frac{1}{\sqrt{d_k}}$ stabilizes the variance to 1, preventing vanishing gradients during training.

4. Deep Reinforcement Learning in Continuous Control (10 marks)

- **(1) On-policy vs. Off-policy (4 marks):**

On-policy: The agent learns the value of the policy that is currently being used to make decisions and collect data (e.g., PPO). Data is discarded after use.

Off-policy: The agent learns the value of an optimal target policy independently of the agent's current exploratory behavior policy (e.g., DQN). This allows the use of a Replay Buffer to reuse past experiences.

- **(2) Continuous Control Advantage (6 marks):**

Value-based methods (like DQN) require finding the maximum Q -value over all possible actions: $\arg \max_a Q(s, a)$. In a continuous action space (like joint torques), finding this global maximum iteratively at every step is computationally intractable. Policy Gradient / Actor-Critic methods directly parameterize the policy $\pi(a|s)$ using a neural network that outputs the parameters of a continuous probability distribution (e.g., mean and variance of a Gaussian). The agent simply samples an action from this distribution, completely avoiding the expensive $\arg \max$ operation.

5. Inductive Biases: CNNs vs. Vision Transformers (10 marks)

- **(1) CNN Inductive Biases (5 marks):**

Locality: CNNs assume that adjacent pixels are highly correlated. This is enforced by using small convolutional kernels (e.g., 3×3) that only process local neighborhoods.

Translation Invariance/Equivariance: The exact location of a feature in an image shouldn't matter. This is enforced by *weight sharing*—sliding the exact same filter across the entire image.

- **(2) ViT Differences and Data Hunger (5 marks):**

ViTs have very little image-specific inductive bias. They flatten image patches into a 1D sequence and use global self-attention, treating every patch as equally distant initially. Because ViTs lack the "built-in" prior knowledge of locality and translation invariance, they must learn these spatial relationships entirely from the data. Consequently, ViTs tend to overfit and perform worse than CNNs on small datasets. However, when trained on massive datasets (e.g., JFT-300M), ViTs bypass the architectural limitations of CNNs, leading to superior representations and state-of-the-art performance.